



AWS Session

Summary – 26-03-2023

- Every cluster has some configuration file called **kubeconfig**.
- **kubectl** is a command for Kubernetes not for eksctl.
- Whenever we create a pod, one of the responsibilities of pod is to launch an app.

```

MINGW64/c/Users/Vimal Daga/Documents/eks_advance_code
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~
$ cd Documents/

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents
$ mkdir eks_advance_code

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents
$ cd eks_advance_code/

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ ls

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ eksctl get cluster
NAME                               REGION    EKSTL CREATED
exciting-wardrobe-1679738433      ap-south-1  True

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ eksctl delete cluster --name exciting-wardrobe-1679738433
2023-03-26 15:27:14 [i] deleting EKS cluster "exciting-wardrobe-1679738433"

true)
g. "C:\Users\Vimal Daga\.kube\eksctl\clusters/attractive-painting-1679824696"
--write-kubeconfig toggle writing of kubeconfig (default true)

Common flags:
-C, --color string toggle colored logs (valid options: true, false, false)
-d, --dumplogs dump logs to disk on failure if set to true
-h, --help help for this command
-v, --verbose int set log level, use 0 to silence, 4 for debugging and 5 for debugging with AWS debug logging (default 3)

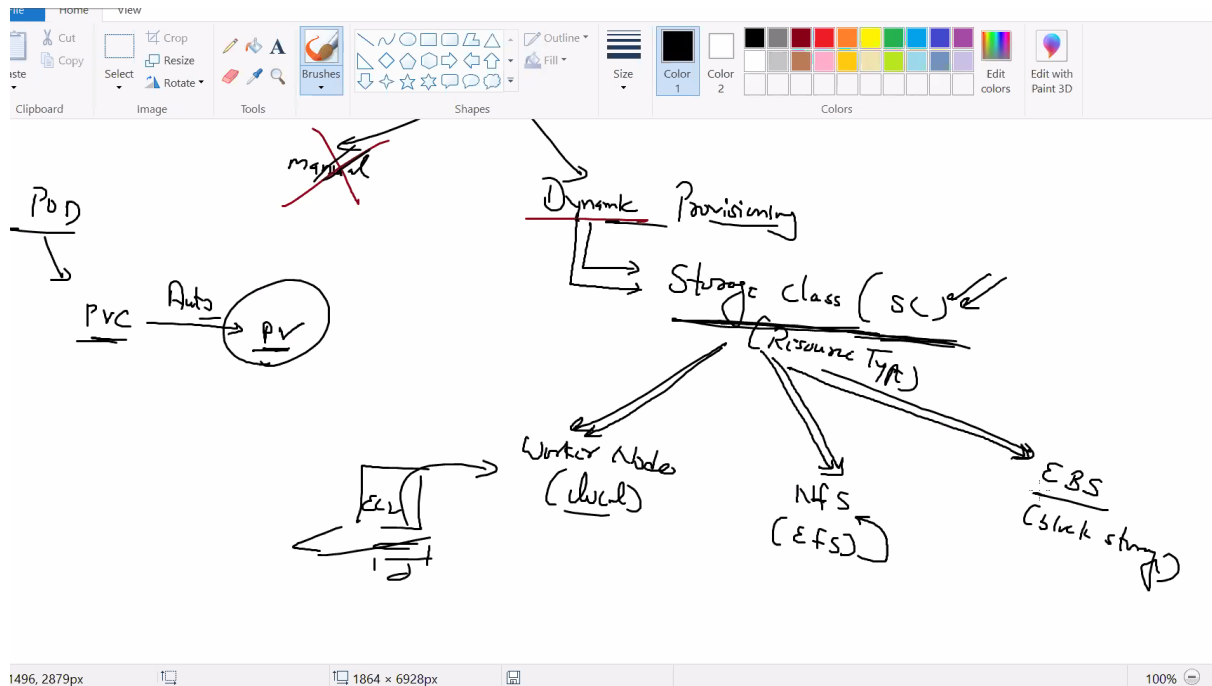
Use 'eksctl create cluster [command] --help' for more information about a command.

For detailed docs go to https://eksctl.io/

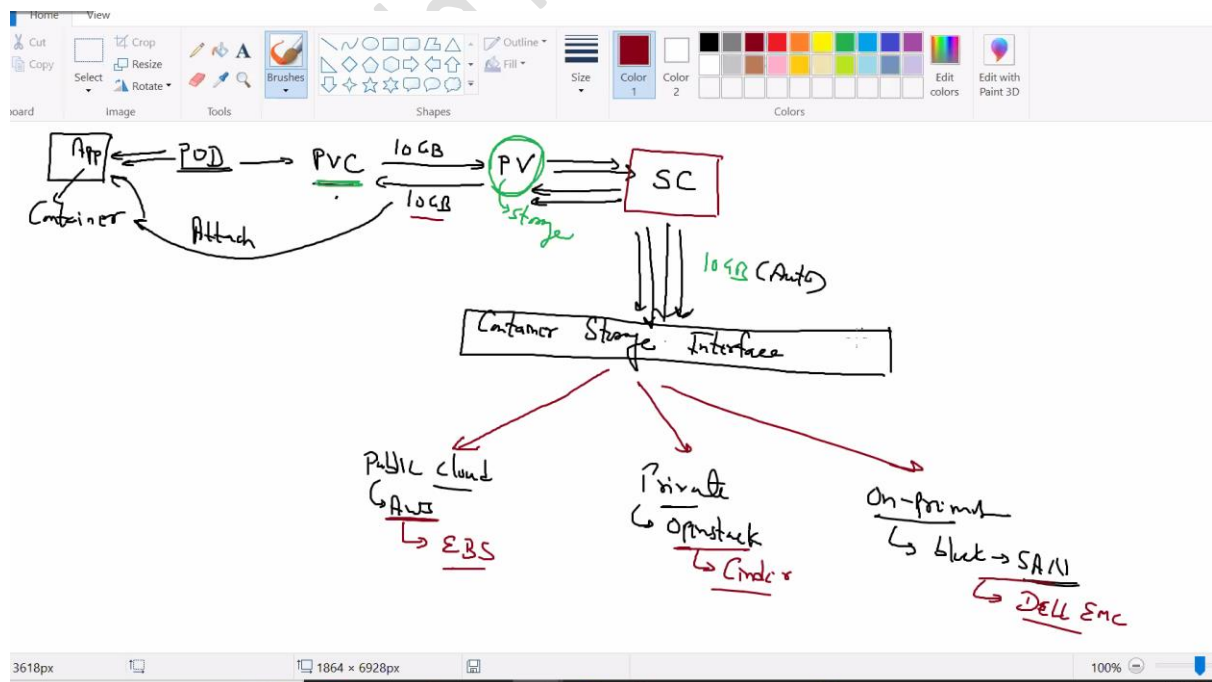
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ eksctl create cluster --name cluster1 --enable-ssm --kubeconfig kube-cluster1.config
  
```

- Application needs storage(permanent/persistent). It is the duty of pod to claim(request) for persistent volume(PV), this concept is called **Persistent volume claim(PVC)**.
- 2 ways to create Persistent Volume (Storage)
 1. Manually
 2. Dynamically

- SC – Storage class help to provision storage dynamically. SC is a resource type.



- Pod is the one who is responsible to run the app.
- Storage class can take storage from:
 1. Public Cloud (AWS- EBS)
 2. Private Cloud (Openstack- Cinder)
 3. On-premises (block-SAN- dell, EMC)



- In k8s, there is one standard interface for the storage. So, if container storage wants to connect with EBS then we need to give a driver/provisioner, So in AWS that driver is known as **CSI Provisioner**.
- Kubernetes only understand PVC, PV, SC If we want Kubernetes resource type take from third party and for that we use CSI.
- Storage class is one that use driver from external provisioners.
- If we run our cluster on k8s with EKS, it will launch K8S and whenever EKS want storage it will connect to storage class (internally Integrated with other services of AWS like EBS, EFS)
- If we want to connect to a cluster then the credentials can be found in kube config file.
- PV is for storage and SC is storage class.
- As soon as we launch pod in EKS and pod need a storage, internally EKS connect with EBS service and launch a new volume and everything is possible because of driver/provisioner.
- Every resource type has annotations which give different responsibilities. In annotations there are keywords like parameters. Annotation is used in all resource type.
- SC need driver called CSI driver. It is a program which runs inside the pod in k8s.
- All internal drivers of k8s is managed in area called **namespace**. One of the namespace is kube-system.
- Whenever EKS launch, it creates role and attach to EKS.
- Configmap is a resource which like a document, maintain config file or maintain file in k8s.
- 2 ways to install driver are
 1. Addons (drivers)
 2. Self-managed addon (some kind of script like k8s scripts)

Standard way is Drivers that is using CSI

The screenshot shows a web browser displaying the Kubernetes Blog article titled "Why CSI?". The URL in the address bar is kubernetes.io/blog/2019/01/15/container-storage-interface-ga/. The page features the Kubernetes logo and navigation links: Documentation, Kubernetes Blog (active), Training, Partners, Community, Case Studies, and Versions. On the left, there is a search bar and a sidebar with a year filter (2023, 2022, 2021, 2020, 2019) and a list of recent articles, including "Kubernetes 1.17: Stability" and "Kubernetes 1.17 Feature: Kubernetes Volume". The main content area is titled "Why CSI?" and contains two paragraphs of text. The first paragraph discusses the challenges of adding support for new volume plugins to Kubernetes before CSI. The second paragraph explains how CSI was developed as a standard to expose arbitrary block and file storage systems to containerized workloads. On the right side of the page, there are social media links for RSS Feeds, GitHub, Twitter (@Kubernetes), GitLab, and a list of community resources like Stack Overflow, Forum, and Kubernetes Slack.

Why CSI?

Although prior to CSI Kubernetes provided a powerful volume plugin system, it was challenging to add support for new volume plugins to Kubernetes: volume plugins were "in-tree" meaning their code was part of the core Kubernetes code and shipped with the core Kubernetes binaries—vendors wanting to add support for their storage system to Kubernetes (or even fix a bug in an existing volume plugin) were forced to align with the Kubernetes release process. In addition, third-party storage code caused reliability and security issues in core Kubernetes binaries and the code was often difficult (and in some cases impossible) for Kubernetes maintainers to test and maintain.

CSI was developed as a standard for exposing arbitrary block and file storage systems to containerized workloads on Container Orchestration Systems (COs) like Kubernetes. With the adoption of the Container Storage Interface, the Kubernetes volume layer becomes truly extensible. Using CSI, third-party storage providers can write and deploy plugins exposing new storage systems in Kubernetes without ever having to touch the core Kubernetes code. This gives Kubernetes users more options for storage and makes the system more secure and reliable.

PRACTICAL

MINGW64:/c/Users/Vimal Daga/Documents/eks_advance_code

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ pwd
/c/Users/Vimal Daga/Documents/eks_advance_code
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ ls
kube-cluster1.config
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ eksctl get cluster
NAME                                REGION      EKSTCTL CREATED
cluster1                            ap-south-1  True
exciting-wardrobe-1679738433        ap-south-1  True
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ ls
kube-cluster1.config
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pods
E0326 16:01:19.625059    14756 memcache.go:238] couldn't get current server API g
roup list: Get "http://localhost:8080/api?timeout=32s": dial tcp [::1]:8080: con
nectex: No connection could be made because the target machine actively refused
it.
E0326 16:01:21.972382    14756 memcache.go:238] couldn't get current server API g
roup list: Get "http://localhost:8080/api?timeout=32s": dial tcp [::1]:8080: con
nectex: No connection could be made because the target machine actively refused
it.
E0326 16:01:24.328932    14756 memcache.go:238] couldn't get current server API g
roup list: Get "http://localhost:8080/api?timeout=32s": dial tcp [::1]:8080: con
nectex: No connection could be made because the target machine actively refused
it.
E0326 16:01:26.692897    14756 memcache.go:238] couldn't get current server API g
roup list: Get "http://localhost:8080/api?timeout=32s": dial tcp [::1]:8080: con
nectex: No connection could be made because the target machine actively refused
it.
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pvc --kubeconfig kube-cluster1.config
No resources found in default namespace.
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ |
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pv --kubeconfig kube-cluster1.config
No resources found
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get sc --kubeconfig kube-cluster1.config
NAME                PROVISIONER          RECLAIMPOLICY  VOLUMEBINDINGMODE
LOWVOLUMEEXPANSION  AGE
gp2 (default)       kubernetes.io/aws-efs  Delete         WaitForFirstConsumer
alse                24m
```


[AWS]

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl describe sc gp2 --kubeconfig kube-cluster1.config
Name: gp2
IsDefaultClass: Yes
Annotations: kubectl.kubernetes.io/last-applied-configuration={"apiVersion":"storage.k8s.io/v1",
"kind":"StorageClass","metadata":{"annotations":{"storageclass.kubernetes.io/is-default-class":"true"},
"name":"gp2"},"parameters":{"fsType":"ext4","type":"gp2"},"provisioner":"kubernetes.io/aws-ebs
","volumeBindingMode":"waitForFirstConsumer"}
,storageclass.kubernetes.io/is-default-class=true
Provisioner: kubernetes.io/aws-ebs
Parameters: fsType=ext4,type=gp2
AllowVolumeExpansion: <unset>
MountOptions: <none>
ReclaimPolicy: Delete
VolumeBindingMode: WaitForFirstConsumer
Events: <none>

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pods --kubeconfig kube-cluster1.config
No resources found in default namespace.

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pods -n kube-system --kubeconfig kube-cluster1.config
NAME                                READY   STATUS    RESTARTS   AGE
aws-node-4l8xs                      1/1     Running   0           28m
aws-node-qgg7w                      1/1     Running   0           28m
coredns-74ccfbd98b-45w5p           1/1     Running   0           36m
coredns-74ccfbd98b-5826h           1/1     Running   0           36m
kube-proxy-6jc1l                    1/1     Running   0           28m
kube-proxy-tkbfd                    1/1     Running   0           28m

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get sc --kubeconfig kube-cluster1.config
NAME                                PROVISIONER             RECLAIMPOLICY   VOLUMEBINDINGMODE   ALLOWVOLUMEEXPANSION
gp2 (default)                      kubernetes.io/aws-ebs   Delete          WaitForFirstConsumer false
38m
```

Gp2 is a storage class created by default.

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get configmap -n kube-system --kubeconfig kube-cluster1.config
NAME                                DATA   AGE
aws-auth                            1       37m
coredns                             1       44m
cp-vpc-resource-controller          0       44m
eks-certificates-controller          0       44m
extension-apiserver-authentication  6       44m
kube-proxy                          1       44m
kube-proxy-config                    1       44m
kube-root-ca.crt                     1       44m

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get configmap aws-auth -n kube-system --kubeconfig kube-cluster1.config
NAME    DATA   AGE
aws-auth 1       37m

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get configmap aws-auth -n kube-system --kubeconfig kube-cluster1.config
NAME    DATA   AGE
aws-auth 1       37m

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl describe configmap aws-auth -n kube-system --kubeconfig kube-cluster1.config
```

Now open IAM and create a role by attaching a permission policy.

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups
- Users
- Roles**
- Policies
- Identity providers
- Account settings

Access reports

- Access analyzer
- Archive rules
- Analizers

IAM dashboard

Security recommendations 1

Add MFA for root user
Add MFA for root user - Enable multi-factor authentication (MFA) for the root user to improve security for this account. [Add MFA](#)

Root user has no active access keys
Using access keys attached to an IAM user instead of the root user improves security.

IAM resources

User groups	Users	Roles	Policies	Identity providers
0	2	12	1	0

What's new [View all](#)

Updates for features in IAM

AWS Account

Account ID: 583457386617
Account Alias: 583457386617 [Create](#)
Sign-in URL for IAM user: <https://583457386617.console.aws.amazon.com/>

Quick Links [My security credentials](#)
Manage your access factor authentication other credentials.

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Roles (12) [Info](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust. [Delete](#) [Create role](#)

Search: eksctl-cluster1-nodegroup-ng-9aa5-NodeInstanceRole-16XLYN14WLI4H 1 match

Role name	Trusted entities
eksctl-cluster1-nodegroup-ng-9aa5-NodeInstanceRole-16XLYN14WLI4H	AWS Service: ec2

Roles Anywhere [Info](#) [Manage](#)

Authenticate your non AWS workloads and securely provide access to AWS services.

Access AWS from your non AWS workloads

X.509 Standard
Use your own existing PKI

Temporary credentials
Use temporary credentials with ease

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Permissions policies (4) [Info](#) [Simulate](#) [Remove](#)

You can attach up to 10 managed policies.

Filter policies by property or policy name and press enter.

Policy name	Type	Description
AmazonEKSWorkerNodePolicy	AWS managed	This policy allows Amazon EKS worker nodes to perform actions on Amazon EKS resources.
AmazonEC2ContainerRegistryReadOnly	AWS managed	Provides read-only access to Amazon ECR resources.
AmazonSSMManagedInstanceCore	AWS managed	The policy for Amazon SSM Managed Instance Core.
AmazonEKS_CNI_Policy	AWS managed	This policy provides the permissions for the Amazon EKS CNI plugin.

Permissions **Trust relationships** **Tags (6)** **Access Advisor** **Revoke sessions**

us-east-1.console.aws.amazon.com/iamv2/home?region=ap-south-1#/roles/details/eksctl-cluster1-nodegroup-ng-9aa5-NodeIn...

Services Search [Alt+S]

☐	AlexaForBusinessGatewayExecution	AWS managed	Provide gateway executi
☐	AmazonElasticTranscoder_ReadOnlyAccess	AWS managed	Grants users read-only a
☐	AmazonRDSFullAccess	AWS managed	Provides full access to A
☐	SupportUser	AWS managed - job function	This policy grants permi
☒	AmazonEC2FullAccess	AWS managed	Provides full access to A
☐	SecretsManagerReadWrite	AWS managed	Provides read/write acco
☐	AWSToTThingsRegistration	AWS managed	This policy allows users
☐	AmazonDocDBReadOnlyAccess	AWS managed	Provides read-only acce
☐	AmazonMQApiFullAccess	AWS managed	Provides full access to A
☐	AWS ElementalMediaStoreReadOnly	AWS managed	Provides read-only perm

Cancel Add permissions

us-east-1.console.aws.amazon.com/iamv2/home?region=ap-south-1#/roles/details/eksctl-cluster1-nodegroup-ng-9aa5-NodeIn...

Services Search [Alt+S]

Identity and Access Management (IAM)

Search IAM

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Permissions

Permissions policies (5) Info

You can attach up to 10 managed policies.

Filter policies by property or policy name and press enter.

☐	Policy name	Type	Description
☐	AmazonEC2FullAccess	AWS managed	Provides full access to Amaz
☐	AmazonEKSWorkerNodePolicy	AWS managed	This policy allows Amazon Ek
☐	AmazonEC2ContainerRegistryReadOnly	AWS managed	Provides read-only access to
☐	AmazonSSMManagedInstanceCore	AWS managed	The policy for Amazon EC2 R
☐	AmazonEKS_CNI_Policy	AWS managed	This policy provides the Ama

Permissions boundary - (not set)

us-east-1.console.aws.amazon.com/iamv2/home?region=ap-south-1#/roles/details/eksctl-cluster1-nodegroup-ng-9aa5-NodeIn...

Services Search [Alt+S]

Identity and Access Management (IAM)

Search IAM

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Permissions

Permissions policies (5) Info

You can attach up to 10 managed policies.

Filter policies by property or policy name and press enter.

☐	Policy name	Type	Description
☐	AmazonEC2FullAccess	AWS managed	Provides full access to Amaz

AmazonEC2FullAccess

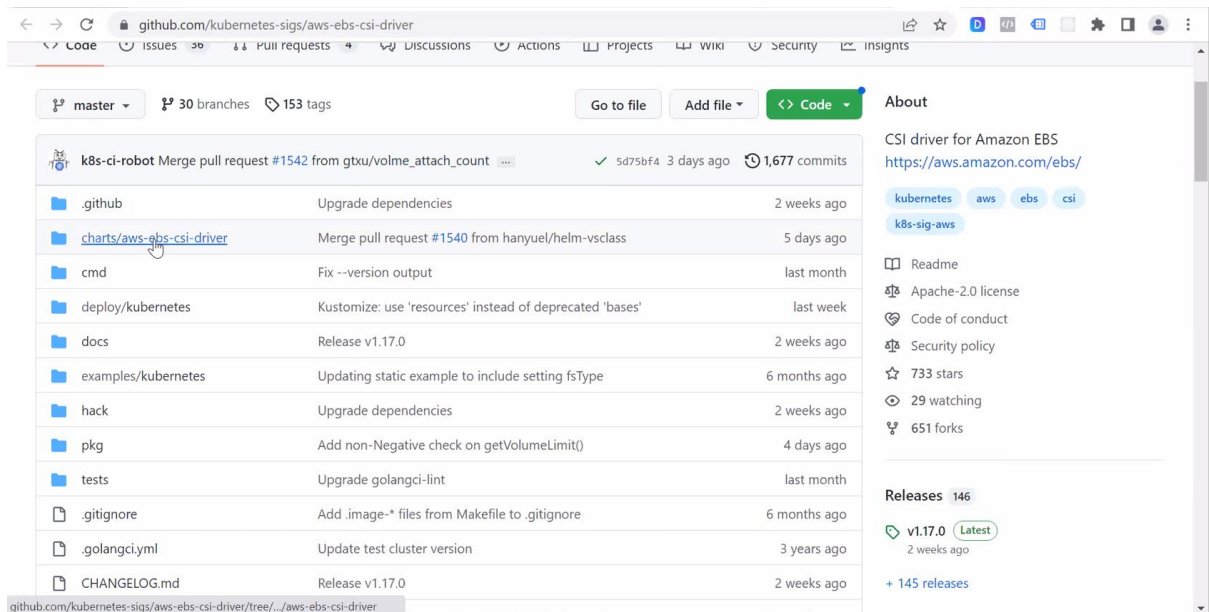
Provides full access to Amazon EC2 via the AWS Management Console.

Copy

```
8      },
9      {
10         "Effect": "Allow",
11         "Action": "elasticloadbalancing:*",
12         "Resource": "*"
13      },
14      {
15         "Effect": "Allow",
16         "Action": "cloudwatch:*",
17         "Resource": "*"
18      },
19      {
20         "Effect": "Allow",
21         "Action": "autoscaling:*",
22         "Resource": "*"
23      }
```

[AWS]

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ eksctl get addon --cluster cluster1
2023-03-26 16:32:59 [i] Kubernetes version "1.24" in use by cluster "cluster1"
2023-03-26 16:32:59 [i] getting all addons
```



github.com/kubernetes-sigs/aws-ebs-csi-driver

master 30 branches 153 tags

Go to file Add file Code

About

CSI driver for Amazon EBS
<https://aws.amazon.com/ebs/>

kubernetes aws ebs csi

k8s-sig-aws

Readme

Apache-2.0 license

Code of conduct

Security policy

733 stars

29 watching

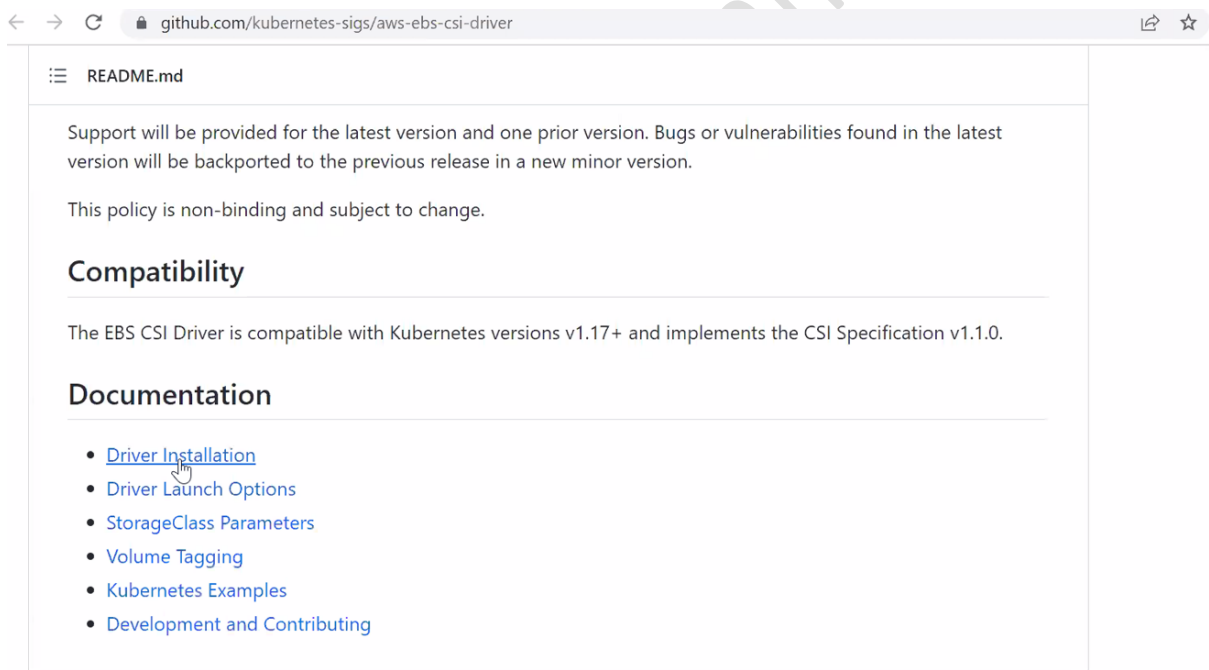
651 forks

Releases 146

v1.17.0 Latest
2 weeks ago

+ 145 releases

File	Commit Message	Time Ago
.github	Upgrade dependencies	2 weeks ago
charts/aws-ebs-csi-driver	Merge pull request #1540 from hanyuel/helm-vsclass	5 days ago
cmd	Fix --version output	last month
deploy/kubernetes	Kustomize: use 'resources' instead of deprecated 'bases'	last week
docs	Release v1.17.0	2 weeks ago
examples/kubernetes	Updating static example to include setting fsType	6 months ago
hack	Upgrade dependencies	2 weeks ago
pkg	Add non-Negative check on getVolumeLimit()	4 days ago
tests	Upgrade golangci-lint	last month
.gitignore	Add .image-* files from Makefile to .gitignore	6 months ago
.golangci.yml	Update test cluster version	3 years ago
CHANGELOG.md	Release v1.17.0	2 weeks ago



github.com/kubernetes-sigs/aws-ebs-csi-driver/tree/.../aws-ebs-csi-driver

README.md

Support will be provided for the latest version and one prior version. Bugs or vulnerabilities found in the latest version will be backported to the previous release in a new minor version.

This policy is non-binding and subject to change.

Compatibility

The EBS CSI Driver is compatible with Kubernetes versions v1.17+ and implements the CSI Specification v1.1.0.

Documentation

- [Driver Installation](#)
- [Driver Launch Options](#)
- [StorageClass Parameters](#)
- [Volume Tagging](#)
- [Kubernetes Examples](#)
- [Development and Contributing](#)

← → ↺ github.com/kubernetes-sigs/aws-ebs-csi-driver/blob/master/docs/install.md

155 lines (130 sloc) 6.88 KB

By default, the driver controller tolerates taint `CriticalAddonsOnly` and has `tolerationSeconds` configured as `300`; and the driver node tolerates all taints. If you don't want to deploy the driver node on all nodes, please set Helm `Value.node.tolerateAllTaints` to false before deployment. Add policies to `Value.node.tolerations` to configure customized toleration for nodes.

Deploy driver

You may deploy the EBS CSI driver via Kustomize, Helm, or as an [Amazon EKS managed add-on](#).

Kustomize

```
kubectl apply -k "github.com/kubernetes-sigs/aws-ebs-csi-driver/deploy/kubernetes/overlays/stable/?ref=release-1.17"
```

Note: Using the master branch to deploy the driver is not supported as the master branch may contain upcoming features incompatible with the currently released stable version of the driver.

Helm

- Add the `aws-ebs-csi-driver` Helm repository.

```
helm repo add aws-ebs-csi-driver https://kubernetes-sigs.github.io/aws-ebs-csi-driver
helm repo update
```

```
$
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl apply -k "github.com/kubernetes-sigs/aws-ebs-csi-driver/deploy/kubernetes/overlays/stable/?ref=release-1.17" --kubeconfig kube-cluster1.config
serviceaccount/ebs-csi-controller-sa created
serviceaccount/ebs-csi-node-sa created
clusterrole.rbac.authorization.k8s.io/ebs-csi-node-role created
clusterrole.rbac.authorization.k8s.io/ebs-external-attacher-role created
clusterrole.rbac.authorization.k8s.io/ebs-external-provisioner-role created
clusterrole.rbac.authorization.k8s.io/ebs-external-resizer-role created
clusterrole.rbac.authorization.k8s.io/ebs-external-snapshotter-role created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-attacher-binding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-node-getter-binding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-provisioner-binding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-resizer-binding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-snapshotter-binding created
deployment.apps/ebs-csi-controller created
poddisruptionbudget.policy/ebs-csi-controller created
daemonset.apps/ebs-csi-node created
csidriver.storage.k8s.io/ebs.csi.aws.com created
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pods -n kube-system --kubeconfig kube-cluster1.config
NAME                                READY   STATUS    RESTARTS   AGE
aws-node-4l8xs                      1/1     Running   0           49m
aws-node-qgg7w                      1/1     Running   0           49m
coredns-74ccfbd98b-45w5p            1/1     Running   0           58m
coredns-74ccfbd98b-5826h            1/1     Running   0           58m
ebs-csi-controller-5b769c467d-h8tmx 6/6     Running   0           34s
ebs-csi-controller-5b769c467d-pbbsf 6/6     Running   0           34s
ebs-csi-node-9fm47                  3/3     Running   0           34s
ebs-csi-node-frv2v                  3/3     Running   0           34s
kube-proxy-6jc1l                    1/1     Running   0           49m
kube-proxy-tkbfd                    1/1     Running   0           49m
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get sc --kubeconfig kube-cluster1.config
NAME        PROVISIONER             RECLAIMPOLICY   VOLUMEBINDINGMODE   ALLOWVOLUMEEXPANSION   AGE
gp2 (default)  kubernetes.io/aws-ebs   Delete          WaitForFirstConsumer  false                  60m
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl describe sc gp2 --kubeconfig kube-cluster1.config
Name: gp2
IsDefaultClass: Yes
Annotations: kubectl.kubernetes.io/last-applied-configuration={"apiVersion":"storage.k8s.io/v1",
"kind":"StorageClass","metadata":{"annotations":{"storageclass.kubernetes.io/is-default-class":"true"},
"name":"gp2"},"parameters":{"fsType":"ext4","type":"gp2"},"provisioner":"kubernetes.io/aws-ebs",
"volumeBindingMode":"WaitForFirstConsumer"}
,storageclass.kubernetes.io/is-default-class=true
Provisioner: kubernetes.io/aws-ebs
Parameters: fsType=ext4,type=gp2
AllowVolumeExpansion: <unset>
MountOptions: <none>
ReclaimPolicy: Delete
VolumeBindingMode: WaitForFirstConsumer
Events: <none>
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pods -n kube-system -l app.kubernetes.io/name=aws-ebs-csi-driver --kubeconfig kube-cluster1.config
NAME                                READY   STATUS    RESTARTS   AGE
ebs-csi-controller-5b769c467d-h8tmx 6/6     Running   0           4m20s
ebs-csi-controller-5b769c467d-pbbsf 6/6     Running   0           4m20s
ebs-csi-node-9fm47                  3/3     Running   0           4m20s
ebs-csi-node-frv2v                  3/3     Running   0           4m20s
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl apply -f https://raw.githubusercontent.com/vimallinuxworld13/aws_eks_code/master/storage_class_add_for_ebs_csi --kubeconfig kube-cluster1.config
storageclass.storage.k8s.io/ebs-sc created
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get sc --kubeconfig kube-cluster1.config
NAME      PROVISIONER          RECLAIMPOLICY   VOLUMEBINDINGMODE   ALLOWVOLUMEEXPANSION
AGE
ebs-sc    ebs.csi.aws.com      Delete          WaitForFirstConsumer false
7s
gp2 (default) kubernetes.io/aws-ebs Delete          WaitForFirstConsumer false
64m
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl describe sc ebs-sc --kubeconfig kube-cluster1.config
Name: ebs-sc
IsDefaultClass: No
Annotations: kubectl.kubernetes.io/last-applied-configuration={"apiVersion":"storage.k8s.io/v1",
"kind":"StorageClass","metadata":{"annotations":{"storageclass.kubernetes.io/is-default-class":"true"},
"name":"ebs-sc"},"provisioner":"ebs.csi.aws.com",
"volumeBindingMode":"WaitForFirstConsumer"}
Provisioner: ebs.csi.aws.com
Parameters: <none>
AllowVolumeExpansion: <unset>
MountOptions: <none>
ReclaimPolicy: Delete
VolumeBindingMode: WaitForFirstConsumer
Events: <none>
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl describe sc gp2 --kubeconfig kube-cluster1.config
Name: gp2
IsDefaultClass: Yes
Annotations: kubectl.kubernetes.io/last-applied-configuration={"apiVersion":"storage.k8s.io/v1",
"kind":"StorageClass","metadata":{"annotations":{"storageclass.kubernetes.io/is-default-class":"true"},
"name":"gp2"},"parameters":{"fsType":"ext4","type":"gp2"},"provisioner":"kubernetes.io/aws-ebs",
"volumeBindingMode":"WaitForFirstConsumer"}
,storageclass.kubernetes.io/is-default-class=true
Provisioner: kubernetes.io/aws-ebs
Parameters: fsType=ext4,type=gp2
AllowVolumeExpansion: <unset>
MountOptions: <none>
ReclaimPolicy: Delete
VolumeBindingMode: WaitForFirstConsumer
Events: <none>
```

```

Vimal Daga@DESKTOP-
$ kubectl describe
Name: gp2
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"storage.k8s.io/v1","kind":"StorageClass",
      "name":"gp2"},
      {"kind":"StorageClass",
      "name":"gp2",
      "volumeBindingMode":"WaitForFirstConsumer"}
  name: gp2
  namespace: storageclass.kubernetes.io
spec:
  provisioner: ebs.csi.aws.com
  parameters:
    AllowVolumeExpansion: true
    MountOptions:
    - fsx
    ReclaimPolicy: Delete
    volumeBindingMode: WaitForFirstConsumer
status:
  events:

```

```
vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ vim mysc.yml
```

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: ebs-sc
provisioner: ebs.csi.aws.com
volumeBindingMode: WaitForFirstConsumer
parameters:
  type: gp3
  fsType: ext4
~
~
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~
~
~
~
~
~
mysticml[+] [unix] (05:29 01/01/1970) 9,15 All
-- INSERT --
```

pg. 11


```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl describe sc ebs-sc --kubeconfig kube-cluster1.config
Name:          ebs-sc
IsDefaultClass: No
Annotations:   kubectl.kubernetes.io/last-applied-configuration={"apiVersion":"storage.k8s.io/v1",
"kind":"StorageClass","metadata":{"annotations":{"name":"ebs-sc"},"parameters":{"fsType":"ext4",
"type":"gp3"},"provisioner":"ebs.csi.aws.com","volumeBindingMode":"WaitForFirstConsumer"}}
Provisioner:    ebs.csi.aws.com
Parameters:    fsType=ext4,type=gp3
AllowVolumeExpansion: <unset>
MountOptions:    <none>
ReclaimPolicy:   Delete
VolumeBindingMode: WaitForFirstConsumer
Events:         <none>
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl apply -f https://raw.githubusercontent.com/vimallinuxworld13/aws_eks_code/master/PVC_ebs-sc --kubeconfig kube-cluster1.config
persistentvolumeclaim/ebs-mysql-pv-claim created
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pvc --kubeconfig kube-cluster1.config
NAME                                STATUS    VOLUME    CAPACITY    ACCESS MODES    STORAGECLASS    AGE
ebs-mysql-pv-claim                  Pending   ebs-sc     4Gi          RWO              ebs-sc          3m29s
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl apply -f https://raw.githubusercontent.com/vimallinuxworld13/aws_eks_code/master/mysql_db_secret.yml --kubeconfig kube-cluster1.config
secret/mysecure created
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get secrets --kubeconfig kube-cluster1.config
NAME      TYPE    DATA    AGE
mysecure  Opaque  2         8s
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl apply -f https://raw.githubusercontent.com/vimallinuxworld13/aws_eks_code/master/mysql_deploy_pvc_ebs.yml --kubeconfig kube-cluster1.config
pod/dbpod1 created
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pvc --kubeconfig kube-cluster1.config
NAME                                STATUS    VOLUME                                CAPACITY    ACCESS MODES
ebs-mysql-pv-claim                  Bound     pvc-8b96cd8e-06ed-4108-a658-6661f16a6b72  4Gi          RWO
ebs-sc                              Bound     pvc-8b96cd8e-06ed-4108-a658-6661f16a6b72  4Gi          RWO
```

The screenshot shows the AWS Management Console interface for the 'Volumes' section. The left sidebar contains navigation links for various AWS services. The main content area displays a table of EBS volumes. The table has columns for Name, Volume ID, Type, Size, IOPS, Throughput, and Snapshot. The table lists several volumes, including 'cluster1-ng-9a...' and 'vol-01812eed45240b7cd'. The 'Create volume' button is visible in the top right corner.

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot
cluster1-ng-9a...	vol-01812eed45240b7cd	gp3	80 GiB	3000	125	snap-06bf46b043eae6e
cluster1-ng-9a...	vol-08923e2b4e96caee	gp3	80 GiB	3000	125	snap-06bf46b043eae6e
-	vol-0af66f522431a978b	gp3	4 GiB	3000	125	-
-	vol-0746107dfa758badf	gp2	10 GiB	100	-	snap-02ac7c56a404c14
-	vol-0479e48cfd0787e29	gp3	8 GiB	3000	125	snap-0fb864c4aa404b7
-	vol-0ee47d64e7746e0f7	gp3	8 GiB	3000	125	snap-0fb864c4aa404b7
-	vol-0bfd608abb58c525e	gp3	8 GiB	3000	125	snap-0fb864c4aa404b7
-	vol-08fde0c4fddc49419	gp3	4 GiB	3000	125	-
-	vol-009048dc337ff029c	gp3	4 GiB	3000	125	-

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Volumes (1/9)

Search

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot
☐	-	vol-0746107dfa758badf	gp2	10 GiB	100	-	snap-02ac7c56a404c14
☐	-	vol-0479e48cfd0787e29	gp3	8 GiB	3000	125	snap-0fb864c4aa404b;
☐	-	vol-0ee47d64e7746e0f7	gp3	8 GiB	3000	125	snap-0fb864c4aa404b;
☐	-	vol-0bfd608abb58c525e	gp3	8 GiB	3000	125	snap-0fb864c4aa404b;
☐	-	vol-08fde0c4fddc49419	gp3	4 GiB	3000	125	-
☐	-	vol-009048dc337f029c	gp3	4 GiB	3000	125	-
☐	cluster1-ng-9a...	vol-01812eed45240b7cd	gp3	80 GiB	3000	125	snap-06bf46b043eae6i
☐	cluster1-ng-9a...	vol-08923e2b4e96caae	gp3	80 GiB	3000	125	snap-06bf46b043eae6i
☒	-	vol-0af66f522431a978b	gp3	4 GiB	3000	125	-

Volume ID: vol-0af66f522431a978b

Details Status checks Monitoring Tags

ap-south-1.console.aws.amazon.com/ec2/v2/home?region=ap-south-1#VolumeDetails:volumeId=vol-0af66f522431a978b

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Details

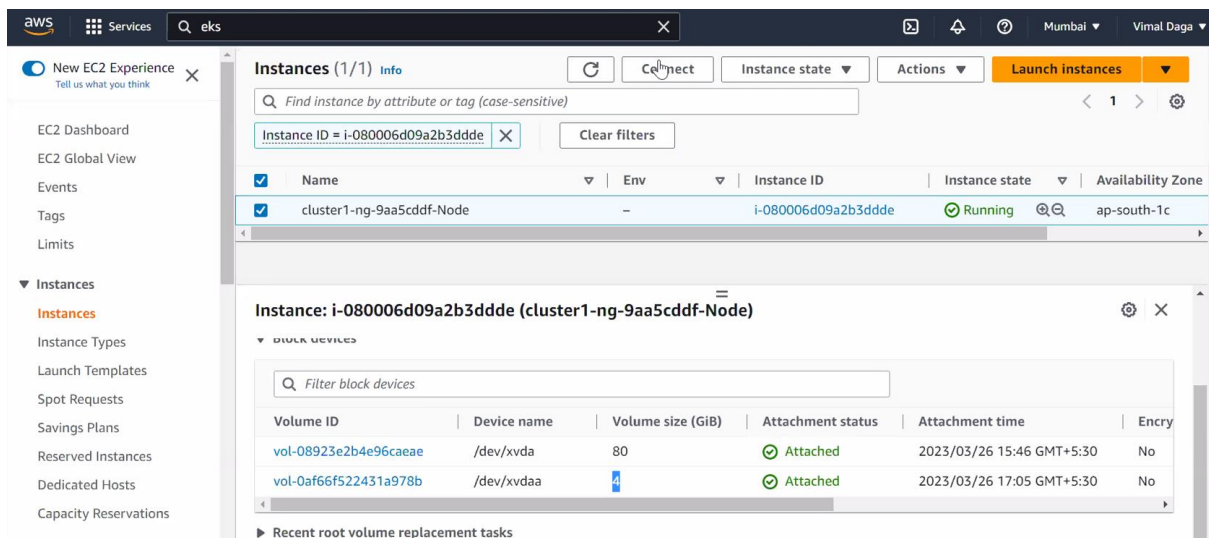
Volume ID vol-0af66f522431a978b	Size 4 GiB	Type gp3	Volume status Insufficient data
AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more	Volume state In-use	IOPS 3000	Throughput 125
Encryption Not encrypted	KMS key ID -	KMS key alias -	KMS key ARN -
Snapshot -	Availability Zone ap-south-1c	Created Sun Mar 26 2023 17:05:29 GMT+0530 (India Standard Time)	Multi-Attach enabled No
Attached Instances i-08006d09a2b3ddde (cluster1-ng-9aa5cddf-Node):	Outposts ARN -		

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pvc --kubeconfig kube-cluster1.config
NAME                                STATUS    VOLUME                                     CAPACITY   ACCESS MODES
ebs-mysql-pv-claim                 Bound     pvc-8b96cd8e-06ed-4108-a658-6661f16a6b72  4Gi        RWO
ebs-sc                              10m

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pv --kubeconfig kube-cluster1.config
NAME                                CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS   CLAIM
pvc-8b96cd8e-06ed-4108-a658-6661f16a6b72  4Gi        RWO             Delete           Bound    default/ebs-mysql-pv-claim
ult/ebs-mysql-pv-claim                 ebs-sc          9s

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pods --kubeconfig kube-cluster1.config
NAME    READY   STATUS    RESTARTS   AGE
dbpod1  1/1     Running   0           2m39s

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$
```



```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pods --kubeconfig kube-cluster1.config
NAME      READY   STATUS    RESTARTS   AGE
dbpod1    1/1     Running   0           5m54s
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl --kubeconfig kube-cluster1.config exec dbpod1 date
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl ex
c [POD] -- [COMMAND] instead.
Sun Mar 26 11:43:10 UTC 2023

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl --kubeconfig kube-cluster1.config exec dbpod1 -- date
Sun Mar 26 11:43:15 UTC 2023
```

- Pod becomes consumer of PVC.
- 1 EBS can connect to only a single pod.
- Multi AZ is not supported by EBS so we can use EFS for this kind of usecase.
- Annotations is for machine It is used as a kind of metadata, k8s gets lots of information from annotation.

Github Link-

https://github.com/vimallinuxworld13/aws_eks_code